

Intro and the division algorithm



Notations:

the natural numbers $\mathbb{N} = \{0, 1, 2, 3, \dots\}$.

the integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$

Definitions 1. An integer a is divisible by

an integer n if there is an integer m such

that $a = mn$. We also say n divides a .

and denote it by $n \mid a$.

Def. 2. An integer a is called even if

there is some integer k such that $a = 2k$.

Otherwise, a is not even.

An integer a is called odd if there is some integer k such that $a = 2k + 1$.

Question 3. Is an integer odd iff not even?

Remark 4 It is important to make it clear what we have assumed when writing a proof.

Otherwise, we will have a one-line proof

that it is obvious that they are the same.

Axiom 5 Every nonempty subset of the

natural numbers has a least element.

Remark ⁶ (1) Note that if we replace $\mathbb{N}$

with $\mathbb{Z}$, the statement is not true.

(2). Also what is hidden in the statement is we have a total ordering on $\mathbb{N}$. If we move to $\mathbb{Q}$, then there is no total ordering.

Theorem 7 For every positive integers a, b

there exists nonnegative integers q and r

with $r < a$ such that

$$b = qa + r$$

Moreover, if $b = q'a + r'$ with $r' < a$,

then $q = q'$ and $r = r'$.

proof. (Existence) Assume $a, b \in \mathbb{N}$, $a \cdot b > 0$.

Consider the set $R = \{ b - na : n \in \mathbb{Z} \} \cap \mathbb{N}$.

Clearly $R \neq \emptyset$ as $b = b - 0 \cdot a \in R$

Then by the Well-ordering principal,

R contains a least element, say $r = b - qa$.

We claim $0 \leq r < a$. If $r \geq a$, then

$r - a \in R$ and $r - a < a$ which

contradicts that r is the least element.

This proves the existence of q and r .

$r \in R$ implies $r \geq 0$. if $b < a$. $q = 0$

if $b \geq a$. $q > 0$

(Uniqueness). Suppose $b = qa + r = q'a + r'$

for some $q, q', r, r' \geq 0$ and $r, r' < a$.

Without the loss of generality, assume $q \geq q'$

(if not, rename q by q' and accordingly)

Then $0 \leq (q - q')a = r' - r < a$, which implies

$$q - q' = 0 \quad \text{i.e.} \quad q = q'.$$

It follows $r' - r = (q - q')a = 0$ i.e. $r = r'$.

Remark: 8 (1) How do we use the assumption $a, b > 0$?

$$(2) \quad \text{if } r, r' < a, \text{ then } |r' - r| < a$$

$$(3) \quad \text{if } k \geq 1, \quad ka \geq a.$$

If we drop the assumption that $b > 0$

the set $R = \{ b - na : n \in \mathbb{Z} \} \cap \mathbb{N}$

can still be showed as nonempty..

But we can't preserve q being nonnegative.

For instance, $-5 = 2 \cdot q + r$ for $0 \leq r < 2$

If $q \geq 0$, $r = -5 - 2q \leq -5$

Theorem: let a, b be integers and $a > 0$

Then exists unique integers q, r

such that $b = qa + r$ with $0 \leq r < a$

proof. (Existence)

Consider the set $R = \{b - na : n \in \mathbb{Z}\} \cap \mathbb{N}$.

If $b > 0$, we are done.

If $b = 0$, still okay for n can be 0.

If $b < 0$, let $n = -N$ for $N > 0$.

then $b - na = b + Na \geq 0$ holds for

sufficiently large N by the Archimedean

property. So R admits a least element

$r = b - qa$. The rest is similar.

Lemma 10 (the Archimedean property)

If x is a real number, then, there

exists a natural number N s.t. $x < N$

Corollary 11 If $x, y \in \mathbb{R}$. $x > 0$.

then $\exists N \in \mathbb{N}$ s.t. $y < xN$

So in particular, $b - na = b + Na \geq 0$

iff $Na \geq -b$ which is true by the Corollary.

Corollary 12. Any integer is odd iff

it is not even.

Exercise :

1. Define $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ for $n \geq k \geq 0$

a) $\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}$

b) $\sum_{k=0}^n \binom{n}{k} = 2^n$

c) compute $\sum_{0 \leq i \leq n} \binom{d+i}{i}$

2. Let n be a fixed positive integer.

The division algorithm defines a function $(-)\bmod n$

$$\mathbb{Z} \rightarrow \mathbb{N} \quad \text{by} \quad b \mapsto r$$

where $b = qn + r$, $0 \leq r < n$

a) describe the preimage of r

for each $0 \leq r < n$

b) Deduce $(-)\bmod n$ determines
a partition of $\mathbb{Z}$.

Remark: $(-)\bmod n$ is a set function
right now but it is also a group or
a ring homomorphism.